Contents

Awesome Mnemonics - Complete Guide

Memory hacks that compress complex workflows into actionable acronyms. Use during incidents, RCAs (root cause analysis), design decisions, or high-stress situations.

Disclaimer: Established frameworks (e.g. 8D, SWOT, PESTEL, RACI, SET) and curated mnemonics for learning and incident use. Sources & References for attribution.

Release v2.5 — 2026-01-25

In PDF and DOCX, the table of contents and in-text links are clickable; use the TOC to jump to sections.

Quick Reference & On-Call Guide

Production incident? → Crisis Response Chain (STOP → TRACE → DEBUG → 8D)
Need quick decision? → Rapid Triage Chain (IDEA → DICE → FATE)

Mobile: Table scrolls horizontally if needed

Situation	Use This	Time
Immediate crisis	IDEA → STOP	2-5 min
Root cause needed	ICEBERG + 5 Whys	30-60 min
Team conflict	BREATHE → PAUSE → WAIT	5-10 min
Strategic planning	PREPARE + SWOT	1-2 hours
System design	SCALE + PESTEL	Planning
Technical debugging	TRACE → DEBUG	Variable
Stress overload	PACE → ARIES → CALM	10-15 min

(If icons don't display: =crisis, =root cause, =conflict, =planning, =design, =debug, =stress.)

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Categorized Index

Lookup by area; when-to-use guidance.

Ops

Incidents, troubleshooting, ops decisions.

Mnemonic	When to Use
STOP	First response to any crisis—immediate stress management (2-5 min)
IDEA	Quick, simple problems requiring fast action (2-5 min)
TRACE	Network/system diagnostics—start with connectivity tests
DEBUG	Code or system analysis when problem definition is unclear
DICE	Identify blockers (Delay, Incompetence, Conflict, External factors) before execution
FATE	Validate resource availability (Funding, Allocation, Time, Expertise) for feasibility
PEST	Identify external threats (Political, Economic, Social, Technological) to solutions

RCA (Root Cause Analysis)

Deep investigation, recurring issues, prevention.

Mnemonic	When to Use
ICEBERG	Complex problems with hidden root causes (30-60 min)
5 Whys	Drill down to root cause—keep asking until systemic failure is found
Ishikawa (Fishbone)	Visual cause–effect analysis; map multiple causes to a problem
8D Approach	Critical incidents requiring formal resolution and prevention

Mnemonic	When to Use
A3 Problem Solving	Structured one-page problem-solving (Toyota); good for sharing and alignment
PADDER	Data-driven problem solving when patterns need identification
PREPARE	Strategic planning for medium complexity problems (1-2 hours)

Systems Design

Architecture, planning, technical decisions.

Mnemonic	When to Use
SCALE	Infrastructure design—Security, Capacity, Automation, Load balancing, Error handling
SWOT	Evaluate options—Strengths, Weaknesses, Opportunities, Threats (1-2 hours)
PESTEL	Strategic planning considering external factors (Political, Economic, Sociocultural, Technological, Environmental, Legal)
SET	Set stakeholder expectations—pick 2: Good, Fast, Cheap

Human Factors

Team dynamics, conflict resolution, and collaboration

Mnemonic	When to Use
RACI	Resolve role confusion—assign Responsible, Accountable, Consulted, Informed
WAIT	Before speaking—ask "Why am I talking?" to choose the right response
BREATHE	First step when tensions rise—regulate emotions before responding
PAUSE	When BREATHE isn't enough—step away and evaluate before acting (5-20 min)
DICE	Check for Conflict as a blocker in team execution
FATE	Validate team Expertise and resource Allocation

Personal Life

Stress, burnout, resilience.

Mnemonic	When to Use
PACE	Immediate stress management—Physical activity, Avoid unhealthy behaviors, Coping skills, Emotional awareness
ARIES	Lifestyle changes for long-term stress reduction (2-4 weeks)
CALM	Build resilience—Confidence, Awareness, Logic, Mindfulness
SHINE	Ongoing practice for sustainable positivity—Stay present, Healthy perspective, Identify positive activities, Nourish relationships, Express yourself

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Who Uses This?

No entries yet — ***be the first.*** Social proof helps enterprise adoption; your org helps others.

One-line PR, e.g. `**[Acme](https://acme.com)** - SRE uses STOP→TRACE→DEBUG`. How.

CLI

`mnemonic` — pipeline and search from `mnemonics-index.yaml`.

Install: From repo: `pip install -r scripts/requirements.txt` then `./scripts/mnemonic`. Package: `pip install git+https://github.com/StewAlexander-com/Awesome-Mne` *Homebrew: not yet.*

Examples: `mnemonic --help` · `mnemonic pipeline` / `pipeline crisis` · `mnemonic search network` · `mnemonic search stress -o json`

`--output:` `table` (default), `json`, `markdown`.

Mnemonic Selection Flowchart

If hard to read: Quick Reference or Index.

```
              Problem?

  Quick?     Complex?     Stress?

   IDEA      PREPARE/      STOP →
    +        ICEBERG+      PACE
   STOP       5 Whys
```

Flowchart (text summary): Quick? → IDEA+STOP · **Complex?** → PREPARE/ICEBERG+5 Whys · **Stress?** → STOP→PACE. Use Quick Reference or Categorized Index.

Table of Contents

Problem Solving Techniques

Start here for systematic approaches to complex problems.

PREPARE

```
P - Prioritize the problem
```

```
R - Research & brainstorm solutions
E - Evaluate available options
P - Plan steps to resolve issue
A - Act on the plan
R - Reflect on results
E - Evaluate and revise plan as necessary
```

When to use: Strategic planning (1–2 hr); medium complexity, multi-stakeholder; needs structure. *Not for outages—use STOP→TRACE→DEBUG first; PREPARE in post-mortem.*

Pitfalls: Analysis paralysis — if past 30 min on Research, switch to IDEA. Skipping Reflect/Evaluate → recurring issues.

Combines well with: RACI, SWOT, ICEBERG

Example: *Infra migration: PREPARE for structure, RACI for roles, SWOT for cloud provider choice.* ### PADDER

```
P - Pinpoint problem
A - Analyze data and look for patterns
D - Develop solution & consider other ways to solve the issue-
try to have more than one option
D - Design action plan
E - Execute action plan & Monitor Results
R - Reevaluate and refine plan as needed
```

When to use: Data-driven; pattern identification. Pairs with 8D D3 (Interim Containment) for quick fixes.

Combines well with: IDEA (simpler), 8D (formal), A3 (one-page PADDER for sharing)

Real-world example: *Recurring server crashes - Pinpoint timing, Analyze logs for patterns, Develop interim solutions (restart service) + permanent fix (increase memory), Monitor effectiveness* ### ICEBERG

```
I - Identify issue(s)
C - Collect data and analyze situation
E - Examine possible (root) causes
B - Brainstorm solutions
E - Execute solution(s)
R - Review, evaluate, and adjust solutions
G - Gather feedback
```

When to use: Complex, deep analysis (30–60 min); surface symptoms hide root causes; escalate from IDEA when complexity grows.

Pitfalls: Too deep on simple (e.g. 5-min password reset): start with IDEA, escalate if needed. Skipping G (feedback): prevents recurrence; complete the cycle.

Combines well with: 5 Whys, Ishikawa (fishbone for Examine), 8D, IDEA (escalate if needed)

Real-world example: *Network performance degradation - Identify slowness, Collect metrics (latency, packet loss), Examine causes (routing changes, bandwidth saturation), Brainstorm solutions, Execute, Review with team, Gather feedback from users* ### IDEA

```
I - Identify problem
D - Develop Solution
E - Execute Solution
A - Assess Solution
```

When to use: Quick, simple (2–5 min); crisis; use STOP first if stressed.

Pitfalls: Escalation: if past 10 min or multiple blockers → ICEBERG or PREPARE. Skipping A: validate; prevents recurrence.

Combines well with: STOP (crisis), PREPARE/ICEBERG (escalate)

Real-world example: *User can't access VPN - Identify (connection error), Develop (reset credentials), Execute (send new password), Assess (confirm access restored)* ### 5 Whys * Keep asking why until root causes are identified

When to use: Root cause analysis; 8D D4; combine with ICEBERG for deep dives.

Pitfalls: Don't stop at symptoms (e.g. "DB slow")—drill to process/systemic failure. Multiple root causes: use 5 Whys per branch.

Combines well with: ICEBERG, Ishikawa (branches, then "why" on each), 8D

Real-world example: *Deployment failures - Why? Pipeline failed. Why? Tests timed out. Why? Database slow. Why? Index missing. Why? Schema change didn't include migration. Root cause: Missing migration validation step*

Ishikawa (Fishbone) Diagram

Cause–effect; map multiple causes to one problem. Pairs with 5 Whys.

```
1. State the problem (head of the fish)
2. Choose categories (e.g. 6 M's: Machine, Method, Material, Manpower, Measurement, Milieu/
3. Brainstorm causes in each category (bones)
4. Drill into "why" for significant causes (use 5 Whys on branches)
5. Identify root causes to address
```

When to use: Multiple causes (5 Whys may miss branches); 8D D4 or ICEBERG Examine; brainstorm across people, process, tech, environment.

Pitfalls: Too many bones: limit branches, focus on likely. Skipping drill-down: use 5 Whys on likely bones.

Combines well with: 5 Whys, 8D D4, ICEBERG, A3 (fishbone on A3)

Real-world example: *Uptime drop - Problem (head): “Services unreachable.” Bones: Method (recent deploy), Machine (high CPU), Manpower (config change). Drill with 5 Whys on “recent deploy” → missing health-check in pipeline. Root cause: CI didn’t run post-deploy checks.*

8D Approach

Industry standard in automotive/manufacturing, adapted for IT incident management

```
D1 - Form a team
D2 - Describe the problem
D3 - Interim Containment Action (the "band-aid")
D4 - Root Cause Analysis & Escape Point(s)
D5 - Permanent Corrective Actions
D6 - Implement & Validate Corrective Actions
D7 - Prevent reoccurrence(s)
D8 - Congratulate your team and close the loop (closure & celebration)
```

When to use: Critical incidents, formal resolution; documentation and prevention; team coordination (D1: RACI).

Pitfalls: Stopping at D3: reach D7 or it recurs. Solo 8D: use D1 (Form a team) and RACI. Bureaucracy: use IDEA or PREPARE for non-critical.

Combines well with: PADDER (D3), 5 Whys+Ishikawa+ICEBERG (D4), A3 (one-page 8D), RACI (D1)

Real-world example: *Data breach incident - Form security response team (RACI roles), Describe scope, Contain (disable compromised accounts), Analyze root cause (5 Whys: phishing → no MFA → insufficient training), Implement MFA, Validate with penetration test, Prevent (mandatory security awareness), Celebrate team response*

A3 Problem Solving

Toyota’s single-page (A3 size) structured problem-solving. Plan–Do–Check–Act on one sheet for clarity and alignment.

```
1. Background & problem statement
2. Current state / gap
3. Goal / target state
4. Root cause analysis (use 5 Whys or Ishikawa)
5. Countermeasures
6. Implementation plan & follow-up (Check)
```

When to use: Share problem and plan on one page; lighter than 8D; recurring/medium severity; socialize ICEBERG or PADDER output.

Pitfalls: Cramming: if it doesn't fit, split or use 8D. Root cause box: use 5 Whys or Ishikawa, not symptoms.

Combines well with: 5 Whys, Ishikawa (root cause box), 8D (A3 summarizes), ICEBERG, PADDER

Real-world example: *Sprint overruns - A3: Background (delivery slipping), Current state (scope creep, no Definition of Done), Goal (predictable sprints), Root cause (5 Whys → no intake prioritization), Countermeasures (backlog refinement + DoD), Plan (next 2 sprints). Share with product and eng leadership on one page.*

5Ps

Poor planning produces pitiful products. — Use as a reminder during planning; don't skip PREPARE.

Problem Analysis

Frameworks for scope and impact.

RACI

Roles and responsibilities in problem-solving.

```
R - Responsible (does the work)
A - Accountable (final approval)
C - Consulted (provides input)
I - Informed (kept updated)
```

When to use: Role confusion; 8D D1 (Form a team); end of WAIT→BREATHE→PAUSE for conflict.

Pitfalls: One "A" per task. Too many "C" slows decisions—only critical input.

Example: *Infra upgrade: R=DevOps, A=Infra Manager, C=Security, I=All devs.* ### PESTEL *External factors that impact a problem or decision.*

```
P - Political
E - Economic
S - Sociocultural
T - Technological
E - Environmental
L - Legal
```

When to use: Strategic planning, external factors; SCALE validation; architecture, compliance.

Real-world example: *Cloud migration planning - Political (vendor lock-in concerns), Economic (cost optimization), Technological (API compatibility), Legal (data sovereignty requirements)*

SWOT

Internal and external factors for a problem or decision.

```
S - Strengths
W - Weaknesses
O - Opportunities
T - Threats
```

When to use: - PREPARE's "Evaluate options" step - SCALE design validation - Strategic decision making (1-2 hours)

Real-world example: *Choosing deployment strategy - Strengths: automated rollback, Weaknesses: longer deployment time, Opportunities: canary testing, Threats: increased complexity* ### SET (Systems Engineering Triangle) *Good, Fast, Cheap: pick 2; the third suffers.* (Fast+Cheap Good, Fast+Good Cheap, Good+Cheap Fast.)

When to use: Set expectations; end of SCALE→SWOT→PESTEL; architecture trade-offs.

Real-world example: *Urgent security patch — Good+Fast not cheap (overtime). Set expectations with leadership.*

SEBoK

Problem Resolution Threats

Blockers before they derail.

DICE

```
D - Delay
I - Incompetence
C - Conflict
E - External factors
```

When to use: IDEA→DICE→FATE triage; blockers before execution; risk in PREPARE phase.

FATE

```
F - Funding
A - Allocation of resources
T - Time
E - Expertise
```

When to use: Resource validation in triage; feasibility (10–30 min); after DICE to validate resources.

PEST

```
P - Political
E - Economic
S - Social
T - Technological
```

When to use: External threats to solutions; how to combat/remove.

Communication & Conflict

Calm, productive in difficult conversations.

BREATHE

```
B - Breathe deeply and slowly
R - Remain rational and listen
E - Empathize with the other person's problem
A - Ask questions to understand
T - Take a break if needed
H - Hold back from reacting
E - Express yourself calmly
```

When to use: First when tensions rise (breathing regulates); WAIT→BREATHE→PAUSE→RACI; if break needed → PAUSE.

Pitfalls: Fake: breathing must be intentional and deep. Weaponizing: don't use dismissively; damages trust.

Combines well with: WAIT, PAUSE, STOP

Real-world example: *Stakeholder disagrees with technical approach in meeting - Breathe deeply (regulate emotions), Remain rational, Empathize with their concerns, Ask clarifying questions, Take 5-minute break if tension escalates, Hold back defensive reactions, Express technical rationale calmly* ### PAUSE

```
P - Put things in perspective
A - Acknowledge your feelings and theirs
U - Understand that you don't have to act/react right away
S - Step Away from the situation
E - Evaluate options and plan before acting
```

When to use: - When BREATHE isn't enough - need physical separation - "Step Away" connects to WAIT philosophy - Combine with STOP for immediate stress de-escalation (5-20 min)

Combines well with: BREATHE (first step), WAIT (listen before speaking), STOP (stress response)

Real-world example: *Heated debate about architecture decision - Put in perspective (not life-or-death), Acknowledge both viewpoints have merit, Don't decide now, Step away for lunch break, Evaluate pros/cons offline, Return with structured comparison*

WAIT

"Why am I troubled / talking?" — Choose the right response; often listening is better.

When to use: Before speaking; first in WAIT→BREATHE→PAUSE→RACI. Use BREATHE to regulate, then WAIT to choose.

Pitfalls: Not "don't respond"—choose effectively. Don't use WAIT to avoid necessary communication or to dodge escalation.

Combines well with: BREATHE, PAUSE

Example: *Accusatory email: Ask "Why troubled?" (ego?) and "Why talking?" (defend or resolve?). BREATHE+PAUSE; respond later with facts.*

Stress & Resilience

Calm and connected under pressure.

PACE

```
P - Physical activity
A - Avoiding unhealthy behaviors
C - Coping skills
E - Emotional awareness
```

When to use: First in PACE→ARIES→CALM→SHINE; immediate stress.

STOP

```
S - Step back
T - Take a deep breath
O - Observe what is happening
P - Pull back and put things in perspective
```

When to use: - First response to ANY crisis - immediate stress management (2-5 min) - Start of STOP → TRACE → DEBUG → 8D crisis response chain - Use with IDEA for quick problem resolution in crisis

Common pitfalls: - Going through motions - Rushing through STOP without actually calming. The "Take a deep breath" must be intentional - pause for 3-5 seconds. - **Stopping at STOP** - Using STOP but not proceeding to next step (TRACE/IDEA). STOP is the foundation, not the solution.

Combines well with: BREATHE + PAUSE (breathing techniques), IDEA (quick problem solving)

Real-world example: *Production alert at 2 AM - Step back (don't panic), Take deep breath (reduce adrenaline), Observe (read alert details), Pull back perspective (assess severity before waking team), Then proceed to TRACE for diagnostics*

ARIES

```
A - Avoid unnecessary stress
R - Relax and take breaks
I - Incorporate physical activity into your routine
E - Eat a healthy diet
S - Sleep well
```

When to use: Lifestyle in PACE→ARIES→CALM; long-term (2–4 weeks).

CALM

```
C - Confidence: Believe in your abilities and strengths
A - Awareness: Stay conscious of your thoughts and feelings
L - Logic: Use rational thinking to overcome doubts
M - Mindfulness: Practice being present and focused
```

When to use: - Builds long-term resilience and confidence - Confidence connects to SHINE (positive self-image) - Use after PACE or ARIES for comprehensive stress management

Combines well with: SHINE, PACE, ARIES

SHINE

```
S - Stay present, in the moment
H - Have a healthy positive perspective
I - Identify and do positive activities
N - Nourish positive relationships
E - Express yourself
```

When to use: End of PACE→ARIES→CALM→SHINE; ongoing positivity; links to CALM.

Infrastructure & Systems Engineering

Infra and DevOps.

TRACE (Network Troubleshooting)

```
T - Test connectivity (ping, traceroute)
R - Review logs and metrics
A - Analyze packet captures
C - Check configurations
E - Escalate with documented evidence
```

When to use: With PREPARE and 8D; T for quick, A for deep; document for 8D D2.

Combines well with: PREPARE, 8D

SCALE (Infrastructure Design)

```
S - Security by design
C - Capacity planning
A - Automation-first
L - Load balancing
E - Error handling/resilience
```

When to use: SWOT for design validation; PESTEL for external factors; SET for trade-offs per component.

Combines well with: SWOT, PESTEL, SET

DEBUG (Code & System Analysis)

```
D - Define the problem (what changed?)
E - Examine error messages/logs
B - Break down into components
U - Understand data flow
G - Generate hypothesis and test
```

When to use: 5 Whys + ICEBERG; start with D (often misstated); with TRACE for network/system.

Combines well with: 5 Whys, ICEBERG, TRACE

Proven Mnemonic Pipelines

Chains combine mnemonics into workflows for high-pressure scenarios; battle-tested.

1. CRISIS RESPONSE CHAIN

STOP → TRACE → DEBUG → 8D

When: Production outages, system failures, critical incidents · **Time:** 1–4 hr

Flow: STOP → TRACE → DEBUG → 8D. *Step details in each mnemonic above.*

Why: STOP stabilizes; TRACE gathers evidence; DEBUG structures analysis; 8D prevents recurrence.

2. CONFLICT RESOLUTION CHAIN

WAIT → BREATHE → PAUSE → RACI

When: Team disagreements, tense meetings, stakeholder conflicts · **Time:** 5–20 min

Flow: WAIT → BREATHE → PAUSE → RACI. *Step details in each mnemonic above.*

Why: WAIT prevents escalation; BREATHE regulates; PAUSE creates space; RACI clarifies roles (often the root cause).

3. ROOT CAUSE INVESTIGATION CHAIN

ICEBERG → 5 Whys → PADDER → RACI

When: Recurring issues, complex systems, post-incident · **Time:** 1–2 hr

Flow: ICEBERG → 5 Whys → PADDER → RACI. *Step details in each mnemonic above.*

Why: ICEBERG structures; 5 Whys drill to root cause; PADDER plans; RACI ensures accountability.

4. STRATEGIC DESIGN CHAIN

SCALE → SWOT → PESTEL → SET

When: Infrastructure planning, architecture reviews, capacity · **Time:** 2–4 hr

Flow: SCALE → SWOT → PESTEL → SET. *Step details in each mnemonic above.*

Why: SCALE sets requirements; SWOT evaluates; PESTEL finds external risks; SET manages expectations.

5. STRESS BURNOUT RECOVERY CHAIN

PACE → ARIES → CALM → SHINE

When: Long-term stress, approaching burnout, lifestyle reset · **Time:** 2–4 weeks (habit formation)

Flow: PACE → ARIES → CALM → SHINE. *Step details in each mnemonic above.*

Why: PACE (immediate); ARIES (lifestyle); CALM (resilience); SHINE (sustainable).

6. RAPID TRIAGE CHAIN

IDEA → DICE → FATE

When: Quick wins, time-critical decisions, feasibility check · **Time:** 10–30 min

Flow: IDEA → DICE → FATE. *Step details in each mnemonic above.*

Why: IDEA (fast frame); DICE (blockers); FATE (resources).

Pipeline Selection Matrix

Your Situation	Pipeline	Key Benefit
System down NOW	STOP → TRACE → DEBUG → 8D	Systematic crisis response
Team conflict escalating	WAIT → BREATHE → PAUSE → RACI	De-escalation + role clarity
Same issue keeps happening	ICEBERG → 5 Whys → PADDER → RACI	Deep root cause + prevention
Designing new infrastructure	SCALE → SWOT → PESTEL → SET	Complete planning framework
Feeling burned out	PACE → ARIES → CALM → SHINE	Comprehensive stress recovery
Need quick decision	IDEA → DICE → FATE	Fast feasibility assessment

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Pro Tips

1. **Don't skip steps** — Each mnemonic builds on the previous.
2. **Document at each stage** — STOP→TRACE→DEBUG; each step feeds the next.
3. **Branch when needed** — If IDEA reveals complexity, switch to ICEBERG → 5 Whys.
4. **Start with STOP** when emotionally activated.
5. **Teach your team** — Shared vocabulary speeds collaboration.

Common Mistakes

Mistake	Problem	Fix	Example
Jumping to solutions	Skip STOP/WAIT when stressed → bad decisions	STOP first in crises; WAIT before reacting	2 AM alert: SSH without STOP → wrong reboot. STOP (30 s) → assess → TRACE.
Wrong pipeline	IDEA won't fix systemic issues	ICEBERG → 5 Whys for recurring	Weekly DB timeouts: IDEA (restart) → returns. ICEBERG+5 Whys → missing pool config → permanent fix.
Incomplete execution	8D stopped at D3 (band-aid)	Always reach D7 (Prevent Reoccurrence)	—
Solo hero	No RACI → no accountability if you're out	RACI in 8D D1 (Form a team)	—
Analysis paralysis	PREPARE→ICEBERG→5 Whys→8D for simple issues	Start with IDEA; escalate if complexity emerges	—

↑ **Quick Reference** · ↑ **Top**

Contributing

Got a mnemonic that's saved you countless times? **Share it!**

Submission criteria: - Must be provable/actionable (not just motivational) - Should cross-reference existing mnemonics where applicable - Include real-world usage context - Keep it memorable (that's the point!)

See CONTRIBUTING.md for details.

Release Versions

v1.8 (2026-01-25) — Offline-ready (TOC, print-optimized). CHANGELOG

ZIP (all formats): Complete · Quick Reference

By format: Complete — PDF DOCX RTF MD · Quick — PDF DOCX RTF MD. *releases/README for details.*

Sources & References

Problem-Solving Methodologies

- **8D (Eight Disciplines)** — Ford Motor Company (1987). *Team Oriented Problem Solving Manual.* Evolved from TQM; in wide use in automotive and aerospace. Wikipedia
- **5 Whys** — Toyota Production System root cause technique; widely adapted across industries.
- **Ishikawa (Fishbone) Diagram** — Ishikawa, K. (1960s). University of Tokyo. Cause–effect diagram; central to Japanese quality control and Toyota. Wikipedia
- **A3 Problem Solving** — Toyota Production System. Single A3-page structured problem-solving (plan–do–check–act); one-page report for alignment. Wikipedia
- **PADDER, ICEBERG, IDEA, PREPARE** — Curated/educational problem-solving mnemonics for this collection.

Strategic & Analysis Frameworks

- **PESTEL** — Aguilar, F. (1967). *Scanning the Business Environment.* Harvard; later extended to PESTLE/PESTEL (Legal, Environmental). Background
- **PEST** — Four-factor variant (Political, Economic, Social, Technological); conceptually from the PESTEL lineage.
- **SWOT** — Humphrey, A. (1960s–70s). Stanford Research Institute (SRI International); developed with Fortune 500 planning research. e.g. Ninety
- **RACI** — Responsibility Assignment Matrix; emerged ~1950s–70s, no single inventor. Wikipedia RAM
- **SET (Systems Engineering Triangle)** — SEBoK, *Guide to the Systems Engineering Body of Knowledge.* SEBoK

Problem Resolution Threats & Triage

- **DICE, FATE** — Educational mnemonics for blockers and resource checks in this collection.

Communication, Stress & Resilience, Infrastructure

- **BREATHE, PAUSE, WAIT, PACE, STOP, ARIES, CALM, SHINE** — Curated for stress management and conflict de-escalation.

- **TRACE, SCALE, DEBUG** — Curated for infrastructure and troubleshooting.

Note: documented; adapted; curated. Mix of established frameworks and educational compilations.